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Connecting the Portenta Max Carrier to The Things Network

This tutorial explains how to connect your Portenta Max Carrier to the Things Network(TTN) using its onboard LPWAN module.

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Overview

This tutorial explains how to connect your **Arduino® Max Carrier**, with an **Arduino® Portenta H7** to The Things Network (TTN) using its onboard LPWAN module. The article will focus on achieving communication between the Max Carrier and an application on TTN.

Goals

- Enable LoRa® connectivity on the Portenta Max Carrier.
- Establish a connection between the Portenta Max Carrier and TTN.

Required Hardware and Software

[Portenta H7](#)

[Portenta Max Carrier](#)

868-915 MHz antenna with
SMA connector

[USB-C® cable \(USB-C® to
USB-A cable\)](#)

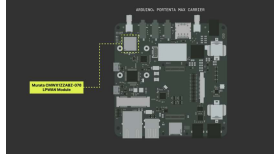
[Arduino IDE 1.8.10+](#),
[Arduino IDE 2.0+](#), or
[Arduino Cloud Editor](#)

[Arduino MKRWAN library](#)

An active account in [TTN](#)

Arduino® Portenta Max Carrier LPWAN Module

The Portenta Max Carrier provides you with an unlimited range of applications, from robotics and medical devices to industrial or automotive applications; the Max Carrier possibilities are endless. One feature that boosts Portenta's Max Carrier possibilities is its **onboard LPWAN module**, the [CMWX1ZZABZ-078](#) from Murata®. LoRa®-based network is a type of Low Power Wide Area Network (LPWAN) designed to connect low power devices to the Internet. It was developed to meet and fulfill Internet of Things (IoT) devices' requirements, such as low-power consumption and low data throughput.



CMWX1ZZABZ-078

LPWAN module in the
Portenta Max Carrier.

For more in-depth
information about LoRa®



and its networks, please
refer to [this](#)
[documentation](#).

Connecting to TTN

Let's start sending data to TTN using the Portenta Max Carrier's LPWAN module. To do this, you will need a TTN account and to be in the range of a public TTN gateway. You can check the [world map](#) of public gateways connected to TTN and see if your region already has a gateway installed. If not, consider installing one!

Check out [this](#) article from
TTN, where you can find



how to buy or build your
gateway to extend TTN
coverage in your region.

To connect your Portenta Max Carrier to TTN, we must follow these steps:

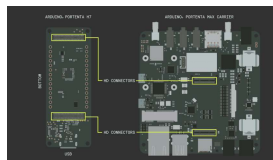
- 1 Hardware setup.
- 2 Software setup.
- 3 Portenta Max Carrier

- 4 Create an application in TTN.
- 5 Send a message to a TTN application.

Let's start!

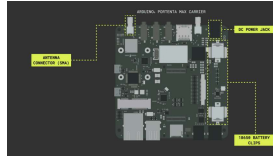
1. Setting up the Hardware

Begin by attaching the Arduino® Portenta H7 board to the high-density connectors of the Portenta Max Carrier, as shown in the image below:



CMWX1ZZABZ-078 module with LoRa® technology on the Portenta Max Carrier.

To power the Murata CMWX1ZZABZ-078 module with LoRa® technology on the Portenta Max Carrier, you can use the **DC power jack** (with a 4.5V to 36V external DC power supply) of the Portenta Max Carrier or a **18650 3.7V Li-Ion battery**, connected to the Portenta Max Carrier battery clips; you can power the module also directly from the USB-C® connector of the Portenta H7 board. **Also, do not forget to attach an 868-915 MHz antenna to the SMA connector (J9) on the Max Carrier.**



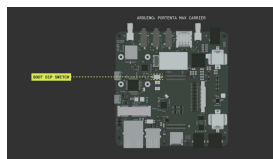
Power sources and
LoRa® antenna
connector on the
Portenta Max Carrier.

Using the LPWAN module
of the Portenta Max
Carrier without an antenna
may damage it. Please, do

i not forget to connect a
suitable antenna to the
dedicated SMA connector
(J9) on the Portenta Max
Carrier.

Now you can connect the
Portenta H7 board to your
computer using a USB-C® cable.

**Don't forget to change the
position of the BOOT DIP
switch (SW1) to OFF ;** otherwise,
you will not be able to program
your Portenta H7 board when
attached to the Portenta Max
Carrier.



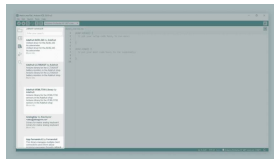
BOOT DIP Switches on
the Portenta Max
Carrier.

2. Setting up the Software

You can use several Arduino

Murata®; we recommend the [MKRWAN library](#), developed by the Arduino Team. The MKRWAN library provides you with the APIs for LoRa® connectivity using the CMWX1ZZABZ-078 module. You can use this library in the Arduino IDE, both [online](#) and [offline](#).

If you are using the online IDE, you don't need to do anything; the library is already installed and ready to be used. If you are using the offline IDE, you must install the library **manually**. Installing the library can be done quickly by navigating to **Tools > Manage Libraries...** and then in the **Library Manager** search for **MKRWAN** library by Arduino; remember to install the latest version of the libraries. You can also access the Library Manager using the left toolbar of the IDE, as shown in the image below:



Library Manager in the Arduino IDE 2.



Currently, there are two versions of the MKRWAN library. We recommend using the MKRWAN_v1 library since MKRWAN_v2 library is still in beta phase.

To use the MKRWAN library with the Portenta Max Carrier, you must define `PORTENTA_CARRIER`

```
1 #define PORTENTA_CARRIER
2 #include <MKRWAN.h>
```

2.1 Updating the LPWAN Module Firmware

The LPWAN module firmware of the Portenta Max Carrier **must be updated** before its first use. This can be done using the example sketch

`MKRWANFWUpdate_standalone` of

the MKRWAN library. You can open this example by navigating to **File > Examples > MKRWAN**.

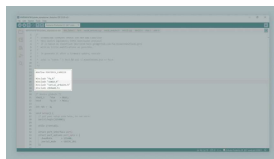
Before uploading the sketch to the Portenta H7 board, open the

`MKRWANFWUpdate_standalone.ino`

file and define

`PORTENTA_CARRIER` before the

library inclusion, as shown in the image below:



Library Manager in the Arduino IDE 2.

Upload the sketch, open the Serial Monitor and wait for the firmware update to finish. You should see a confirmation message when the process is done.

3 Provisioning the

Carrier

Device provisioning is a process comparable to bank card numbering. Let's think about bank cards; bank cards numbers start with a six-digit vendor ID number that indicates who allocated and controls the card's security; the remaining digits are unique numbers associated with a specific card. Devices with LoRa® capabilities have a similar system; the **Join Server Unique Identifier** (usually referred to as `JoinEUI`) is a number that manages the security and authorizes the device in a network, while the **Device Unique Identifier** (usually referred to as `DevEUI`) is a unique number that identifies the device. The `JoinEUI` and `DevEUI` are required to send information to TTN; the `JoinEUI` number is provided by the network (in this case TTN) while the `DevEUI` is provided by the manufacturer of the device's LPWAN module.

The following sketch lets you find out what is the `DevEUI` of your Portenta Max Carrier:


```

1  #define PORTENTA_CARR
2  #include <MKRWAN.h>
3
4  auto region = US915;
5
6  LoRaModem modem(Seria
7
8  void setup() {
9      Serial.begin(115200
10     while(!Serial);
11
12     if(!modem.begin(reg
13         Serial.println("Fa
14         while(1) {}
15     }
16
17     Serial.print("Your
18     Serial.println(modem
19 }
20
21 void loop() {}

```

The only line you may need to change before uploading the code is the one that sets the frequency:

```
1 auto region = US915;
```

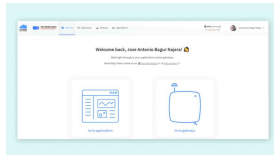
Set the frequency designator according to your country. You can find more information about frequency plans definitions used in TTN [here](#). After you upload the sketch to your Portenta H7 board, you should see the Portenta Max Carrier `DevEUI` in the Arduino IDE Serial Monitor as shown below:

```
1 Your Portenta Max Carr
```

Now, let's use the `DevEUI` number from your Portenta Max Carrier to create an application in TTN.

3. Creating an Application in TTN

To send information to TTN, first we need to **create an application and register a device with it**. Navigate to TTN portal and sign in; after signing in, click on **Create an application**. If you already created an application, click on **Go to applications**.



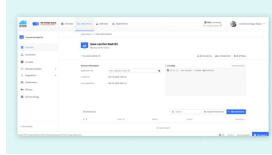
TTN console welcome page.

Now click on **Create an application**. You will need to add the following information:

Owner: the person or organization that owns the application.

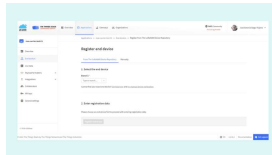
Application ID: a unique identifier for your application (must be lowercase and without spaces).

Complete both fields and click on **Create application**. Now you will be redirected to the application dashboard that shows information of the newly created application.



Application dashboard
in TTN.

Now, scroll to **End devices** in the left toolbar and then click on **Add end device**; a registration page for end devices will open.



Registering an end
device in TTN.

On the registration page, click on **Manually**; you will have to add the following information for your Portenta Max Carrier:

Frequency plan: choose a region according to your country.

LoRa® technology version: 1.0.2.

Regional Parameters version: 1.0.2.

Click on Show advanced activation, **LoRa® class and cluster settings** and choose:

Activation mode: Over the air activation (OTAA).

Additional LoRa® class capabilities: None (class A only).

Network defaults: Use network's default MAC settings.

with the following information:

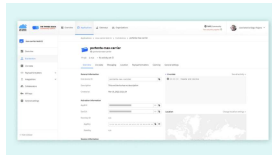
DevEUI: fill it with the `DevEUI` number of your Portenta Max Carrier you found **before**.

AppEUI: fill it with zeros or enter your own.

AppKey: generate one or enter your own.

Device ID: A custom identifier for your board, must be lowercase and without spaces.

Click on **Register end device**; this will take you to a **Device Overview** page where you will see all the information related to the device. Now, you will use some of this information with your Portenta Max Carrier to send data to TTN.



Device overview in TTN.

4. Sending Data to an Application in TTN

Now, let's start sending information to TTN. The following sketch enables you to join and send data to TTN using the Over the Air (OTAA) device activation method. You will need to define your TTN application's AppEUI and AppKey numbers in the `arduino_secrets.h` file before uploading the code:

```
1  #define PORTENTA_CAR
2  #include <MKRWAN.h>
3  #include "arduino_se
4
5  auto region = US915;
6
7  LoRaModem modem(Seri
8
9  void setup() {
10     Serial.begin(11520
11     while (!Serial);
12     Serial.println(F("
13
14     if (!modem.begin(r
15         Serial.println(F
16         while (1) {}
17     };
18
19     Serial.print(F("Yo
20     Serial.println(mod
21
22     if (modem.version(
23         Serial.println(F
24         Serial.println(F
25     }
26
27     Serial.print(F("Yo
28     Serial.println(mod
```

The `arduino_secrets.h` file which contains AppEUI and AppKey numbers for the TTN applications can be defined as following code. AppEUI and AppKey information is found on the End device page of the TTN Application.

```
1  String appEui = "xxxxxx
2  String appKey = "xxxxxx
```

If data was sent correctly to TTN, you should see the following messages in the Arduino IDE

In your device overview dashboard on TTN, you should see changes in data activity every time a new message is sent.

Conclusion

You have now successfully configured and used the onboard LPWAN module of your Portenta Max Carrier. You also have learned how to correctly setup a TTN application and enable LoRa® connectivity between a TTN application and the Portenta Max Carrier.

Next Steps

Scale up the usage of Portenta Max Carrier by using its additional peripherals and turning them into interesting industrial-grade projects, taking advantage of LoRa® connectivity.

Troubleshooting

While working on the sketch or when tried to upload the sketch, the Arduino IDE might show some errors preventing to proceed on the development. You can try the following troubleshooting tips to solve the commonly known issues:

If the sketch upload process fails, check if your Portenta H7 is in bootloader mode. To put the Portenta H7 into

verify that the green LED is waving. After this, you can try re-uploading the sketch.

Check the position of the BOOT DIP switch of the Portenta Max Carrier. If the Portenta H7 gets into bootloader mode immediately after power-on, including when connected via USB-C®, change the position of the BOOT DIP switch to OFF.

If the Arduino IDE fails to compile the sketch, check if you have defined `PORTENTA_CARRIER` before the MKRWAN library inclusion.

Trademark

Acknowledgments

LoRa® is a registered trademark of Semtech Corporation.

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